FinGPT: a retrospective of three LoRA experiments

Fall-2026 Assignment1 | Validation and authorized final-test research report

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Scope. This report retains the original data and reference answers. It evaluates Qwen2.5-7B-Instruct, DeepSeek-R1-Distill-Llama-8B and Llama-3.1-8B BASE, each before and after one epoch of 8-bit LoRA training. It describes observed failures rather than claiming a repaired forecasting dataset.

All three adapters completed 990 training examples and 248 optimizer steps. The same 240 validation identities were used for checkpoint selection and generation. The six model/adapter variants produced 1,440 original-prompt outputs. The frozen final-test protocol subsequently produced 1,800 outputs on 300 identical identities, after real user authorization. An additional 48 HITL prompt-comparison outputs are a separate development experiment.

What the evidence supports

The optimization pipeline works and validation cross-entropy decreases. For the original Qwen and DeepSeek experiments, normalized direction accuracy reaches 145/240 after training, equal to the always-up baseline. Formatting and overlap with reference text improve without establishing reliable forecasting value. These validation results and the separate final-test results below use explicitly named parsers and denominators.

Deliverable	Current boundary
Training and validation	Three trained adapters; complete 240 base + 240 adapter outputs per model
HITL	Actual human claim feedback retained; AI-generated prompt revision evaluated; final human review pending
Final test / publication	All 1,800 test outputs complete;20/280 exposure groups separate. User accepted the materials and authorized publication/submission; README records delivery links

The evaluation is retrospective: data availability was not independently established, some financial headers postdate the prediction origin, and the original DeepSeek supervision leaves an open reasoning boundary. These limits remain after disclosure.

1. Data, environment and training

The frozen FinGPT Dow30 training split has 1,230 rows across 41 prediction weeks. The earlier 33 weeks form train 990; the final eight weeks form validation 240. A prediction week is not split across partitions. Training labels contain 542 up and 448 down; validation has 145 up and 95 down. Dataset labels and reference answers are retained, not independently reconstructed market returns.

Item	Executed setting
Hardware / OS	One remote RTX4090, 24GB, Ubuntu24.04.1; Windows client via SSH; server data volume and local D: evidence
Runtime	Python3.12.3; torch2.7.0a0+7c8ec84dab.nv25.3 (CUDA build12.8); transformers4.51.3; PEFT0.15.2; bitsandbytes0.47.0; accelerate1.7.0; datasets3.6.0
Optimization	Frozen 8-bit base; LoRA r=8, alpha=16, dropout=0.05, q_proj/v_proj; microbatch 1, accumulation 4, peak LR 2e-4; cosine with warmup 0.03; seed 42
Targets	4096-token training context; full references plus EOS; prompt labels -100; no answer clipping
Selection	Full 240 token-weighted validation cross-entropy at steps 0/124/248; select the smaller loss at 124/248, earlier on a tie; selected 248 for each model
Inference	Greedy; context 6144; max new tokens 1920. Qwen batch 1; DeepSeek batch 16/chunk 512; Llama batch 8/chunk 512

The single-GPU Trainer path replaces the starter DeepSpeed example. LR 2e-4 and accumulation 4 differ from course recommendations; reported settings reflect execution. Existing caches were reused and verified; no base weights are included in deliverables.

Qwen and DeepSeek use their native instruction templates. Llama is the BASE model, with native BOS and a plain prefix: source SYSTEM, two newlines, canonical full input, two newlines. It has no chat template. This architecture/prefix difference limits cross-model interpretation.

2. Training evidence and quantitative evaluation

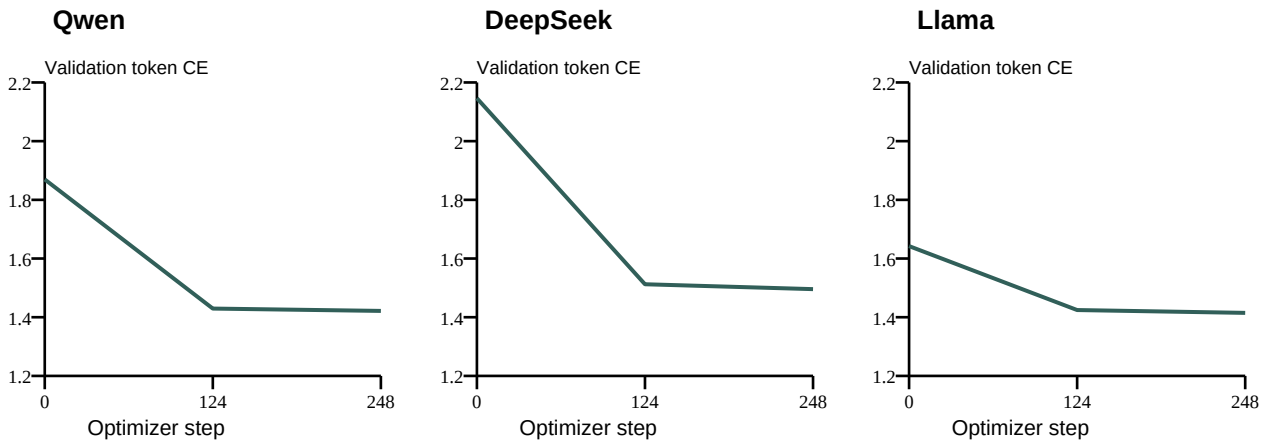

Figure 1. Full validation token-weighted cross-entropy at 0/124/248. Absolute losses are not comparable across tokenizers or model families. This is a traceable post-run plot, not a claimed WandB execution.

Variant	Course valid n	Course MSE	+0.5 MSE, same n	Common direction
qwen base	190/240	11.4842	6.2947	117/240; valid 229
qwen adapter	240/240	7.4292	6.3458	145/240; valid 240
deepseek base	0/240	N/A	N/A	118/240; valid 216
deepseek adapter	240/240	7.5083	6.3458	145/240; valid 240
llama base	190/240	7.4026	6.5474	2/240; valid 4
llama adapter	240/240	7.1375	6.3458	144/240; valid 240

Llama BASE contains 196/240 Up/Down placeholders. Among its 190 course-valid outputs, all 190 parse as up and 178 have zero margin. This is template acceptance, not meaningful forecasting; its course bin_acc equals always-up on that subset. Course MSE measures encoded references, not verified returns. DeepSeek course-valid 0 denotes incompatible layout, not missing outputs. Each baseline uses precisely the same valid subset.

The common direction variant uses the last standalone Prediction line after markdown removal; ambiguous or missing directions count as wrong over all 240. DeepSeek native base requires exactly one closed think boundary. Always-up is 145/240. Historical normalization used broader heading/prediction recognition: Qwen125-to145 and DeepSeek107-to145; the standalone-line rule gives Qwen117-to145 and DeepSeek118-to145. Both variants remain separate; neither adapter beats always-up.

Llama BASE has 65/240 outputs with multiple Prediction lines, versus 0/240 for its adapter. BASE first-line sensitivity: correct 2/240, valid 4; declared last-line rule: correct 2/240, valid 4. First/last disagree in 0 outputs. Repeated or continued text can contain later prediction-looking lines; these counts do not establish answer quality.

Only eight validation weeks are available; rows within a week are related. Validation also selected the checkpoints. These development scores do not establish significance or future-time generalization.

Balanced direction accuracy gives each class equal weight. For trained adapters under the common rule: qwen 52.7%; deepseek 54.5%; llama 55.1%. Always-up gives 50.0%. These modest descriptive differences do not by themselves establish a reliable advantage over future weeks.

3. Reference overlap, completion and runtime

Variant / course n	Positive R1/R2/RL	Concerns R1/R2/RL	Analysis R1/R2/RL
qwen base / 190	0.387/0.120/0.249	0.341/0.088/0.215	0.398/0.093/0.195
qwen adapter / 240	0.462/0.186/0.292	0.425/0.160/0.275	0.438/0.134/0.227
deepseek base / 0	N/A	N/A	N/A
deepseek adapter / 240	0.458/0.184/0.289	0.419/0.155/0.272	0.448/0.141/0.231
llama base / 190	0.050/0.012/0.033	0.026/0.005/0.017	0.075/0.013/0.045
llama adapter / 240	0.474/0.196/0.300	0.425/0.158/0.274	0.439/0.134/0.225

ROUGE F1 measures text overlap within the course-valid pairs. It does not measure factual grounding or financial utility. Different valid denominators must not be treated as matched before/after samples.

Variant	EOS / caps	Batch	Seconds / output*	Peak allocated GiB
qwen base	240 / 0	1	34.79	9.24
qwen adapter	240 / 0	1	47.49	9.25
deepseek base	235 / 5	16	19.46	19.75
deepseek adapter	240 / 0	16	5.30	16.68
llama base	120 / 120	8	36.59	14.19
llama adapter	240 / 0	8	8.92	12.51

*Qwen uses single-answer latency including 30 historical reused outputs; batched DeepSeek and Llama seconds/output are whole-batch time divided by batch size. PyTorch allocated peak differs from reserved and nvidia-smi memory. Batch sizes differ: this is not a same-serving-parameters speed comparison.

EOS completion, output structure and factual accuracy are separate. Original DeepSeek base closes think 240/240; its legacy adapter closes 0/240 despite 240 EOS outputs. The adapter target changes the boundary of the reasoning task. A shorter answer cannot be interpreted as equal-quality reasoning acceleration.

4. Frozen final test: direction and encoded-margin scores

The accepted protocol was frozen before access. The user authorized execution through a direct reply to the frozen-scope approval question; the exact message is retained in the authorization record. The first authorized read was recorded at 2026-10-12T18:50:38.117965+00:00. Model/checkpoint/metric choices were retained, with no new training, test-driven prompt change, filtering or parameter search.

The source has 300 rows across ten prediction weeks, with 137 up and 163 down labels. Twenty rows had been inspected previously. The other 280 are reported separately and are not claimed fully pristine. These are dataset-label and generated-reference comparisons; realized returns and full point-in-time availability were not independently established.

Variant	Course valid	Course MSE	+0.5 MSE, same n	Common direction; BA
qwen base	208/300	11.2356	7.1827	161/300; v296; 0.5352
qwen adapter	297/300	8.7475	7.6465	142/300; v300; 0.5008
deepseek base	0/300	N/A	N/A	130/300; v265; 0.4541
deepseek adapter	297/300	8.5724	7.6465	148/300; v300; 0.5134
llama base	220/300	9.3727	8.3000	2/300; v4; 0.0073
llama adapter	297/300	8.5152	7.6465	150/300; v300; 0.5166

The common rule uses the last standalone Prediction line; missing or ambiguous outputs are wrong. DeepSeek native base requires exactly one closed think marker before final-answer extraction. Always-up and the frozen train-majority-up baseline are 137/300. v is valid direction count; BA is balanced accuracy over all 300, with invalid outputs wrong. The constant +0.5 MSE uses precisely the same course-valid subset. Counting the observed down labels gives 163/300 for constant-down, above every variant (best: Qwen base 161); this is a post-hoc label-distribution fact, not a pre-declared or newly selected primary baseline.

Variant / group n	Direction correct / valid	Balanced accuracy	Always-up
qwen base / 20	12/20; valid20	0.6061	11/20
qwen base / 280	149/280; valid276	0.5307	126/280
qwen adapter / 20	12/20; valid20	0.5758	11/20
qwen adapter / 280	130/280; valid280	0.4964	126/280
deepseek base / 20	10/20; valid17	0.4747	11/20
deepseek base / 280	120/280; valid248	0.4524	126/280
deepseek adapter / 20	8/20; valid20	0.3838	11/20
deepseek adapter / 280	140/280; valid280	0.5238	126/280
llama base / 20	0/20; valid0	0.0000	11/20
llama base / 280	2/280; valid4	0.0079	126/280
llama adapter / 20	7/20; valid20	0.3283	11/20
llama adapter / 280	143/280; valid280	0.5299	126/280

Each variant lists the previously inspected 20 before the other 280. Weekly dependence limits forecasting claims. All raw outputs and 18 metric groups are retained.

5. Final-test overlap, output boundaries and serving

Variant / course n	Positive R1/R2/RL	Concerns R1/R2/RL	Analysis R1/R2/RL
qwen base / 208	0.399/0.125/0.256	0.351/0.096/0.221	0.400/0.096/0.200
qwen adapter / 297	0.453/0.178/0.284	0.430/0.158/0.276	0.443/0.133/0.230
deepseek base / 0	N/A	N/A	N/A
deepseek adapter / 297	0.453/0.178/0.285	0.425/0.156/0.273	0.444/0.140/0.231
llama base / 220	0.043/0.013/0.028	0.020/0.005/0.014	0.065/0.011/0.040
llama adapter / 297	0.461/0.183/0.287	0.437/0.164/0.281	0.441/0.136/0.226

Course raw scores use unchanged generated strings. The DeepSeek base final-extracted parser is an additional separate variant, retained in the JSONs with its own valid_count and same-subset baseline. ROUGE measures overlap, not evidence grounding; different valid subsets are not matched populations.

Variant	EOS / caps	Actual batch sizes	Seconds / output*	Peak allocated GiB
qwen base	300 / 0	12/16	4.43	12.11
qwen adapter	300 / 0	12/16	5.63	12.11
deepseek base	297 / 3	12/16	19.68	17.77
deepseek adapter	300 / 0	12/16	6.28	15.37
llama base	164 / 136	4/8	37.90	13.50
llama adapter	300 / 0	4/8	9.21	12.04

*Test timing is amortized batch time. All variants use greedy generation, full 6144 context,1920 new-token budget and512-token chunk prefill. Default batches are Qwen16, DeepSeek16 and Llama8; final remainder sizes are recorded above. Three independent sequential model processes hold one shared GPU lock. This is not a same-serving-parameters model-speed comparison.

Input-length-only fallback was frozen before access: an original batch containing any prompt above its validation-derived threshold would use batch 1 for every row. All 300 prompts fit the hard 4224-token limit. Actual over-envelope rows: {'qwen': 0, 'deepseek': 0, 'llama': 0}. Base and adapter share identical order and batch partition. Qwen test batch16 differs from its primary validation batch1; an earlier validation serving probe produced0/32 identical texts but31/32 identical parsed directions, so serving sensitivity remains disclosed.

Llama BASE test outputs contain 234/300 Up/Down placeholders. All 220 course-valid pairs parse as up, with 208 zero margins. This is template acceptance, not a meaningful prediction. Generated-only parsing accepts 223 answers; the paired course score uses 220 because 3 references are invalid.

Three test references fail the fixed course parser, so paired course-valid scores have a 297 ceiling. All three adapters have 300 individually parseable answers and 297 valid pairs. Every input and raw answer remains in the300-row evaluation; no row was removed or repaired.

The adapters predict up on 245/219/207 rows for Qwen/DeepSeek/Llama. Their all300 balanced accuracies are 0.5008/0.5134/0.5166; exceeding the frozen always-up137 is not evidence of forecasting skill. Qwen adapter 142 is below its base 161. The test majority is down, unlike the frozen training majority up 542/990.

Qwen and DeepSeek have no multiple Prediction lines. Llama BASE has 61, with 1 first/last disagreement. DeepSeek base has 299 exactly-one-closed think outputs; its adapter has 0. Closed think and EOS are distinct. Original unclosed adapter supervision remains unchanged.

All 1,800 generated token-count/hash and complete boundary receipts were checked. The worker re-decoded tokens using each actual loaded tokenizer before writing. Local CPU checks reproduced all 18 metric groups. Claude Opus5.5 High independently reproduced 900 prompts, 1800 decoded outputs and all 18 metric groups, with 0 actual-run defects. This revision implements its verified text corrections; the auditor explicitly requires no new review round.

6. Human feedback and the matched prompt experiment

The user reviewed 18 selected claims in HON, KO and TRV: eight supported, six partly supported and four uncertain, plus one quick-review note. An earlier partial full-answer review is retained separately. The AI second review disagreed on eight claim labels; original human decisions were not overwritten.

The inference revision adds numbered evidence and AI-authored guidance against vague attention claims, unsupported company attribution and unbenchmarked ratio judgments. It uses unchanged adapters. Three diagnostic cases plus three seeded random additions were frozen before new generation. This six-case development design does not estimate a representative error rate.

Variant	Fresh original direction	Revised direction	Revised citation markers
Qwen base	1/6; valid6	1/6; valid4	5/6
Qwen adapter	4/6; valid6	3/6; valid6	0/6
DeepSeek base	2/6; valid5	3/6; valid6	3/6
DeepSeek adapter	3/6; valid6	4/6; valid6	0/6

Always-up is 4/6 for these cases. Per-model serving settings are matched within each fresh original/revised pair; they differ from the historical DeepSeek batch 16 run. No historical text was substituted into the 48-output experiment.

AI reviewed 24 matched pairs. Some individual errors disappeared while others persisted or appeared. Citation markers sometimes pointed to the wrong financial fields. The revision did not reliably fix grounding; it is retained as a negative optimization result, separate from the primary original-prompt evaluation.

No human rating of the fresh 48 outputs has been recorded. The feedback loop is not marked complete. The user requested AI-first triage; only genuinely uncertain judgments are escalated. AI review and independent AI audit are not human approval.

Analysis depth, clarity and professionalism

In the six previously selected cases, Llama BASE mostly supplies empty templates; HON also continues into unsupported future-period news and a price absent from its input. The trained adapter gives clearer organized analyses with grounded details such as HON expansion and TRV EPS. It still calls financial ratios high or healthy without supplied benchmarks, assigns an unnamed dividend list to GS and an unnamed long-haul list to MSFT, and treats a Vita Coco competitor narrative as a positive development for KO. This is improved presentation with remaining evidence gaps, not demonstrated financial forecasting skill.

The Qwen/DeepSeek matched-case review likewise finds fluent structure with uneven company attribution and causal justification. All these qualitative judgments are AI assessments of selected cases; base continuation excerpts were not exhaustively fact-checked, and no new human verdict was substituted.

7. Limitations, reproducibility and remaining work

Information timing. Financial headers in 61 training and 3 validation rows postdate the prediction origin by 1-6 days. All 1,230 rows lack complete independent availability evidence. This does not prove every row leaked. Excluding 3 validation rows does not repair 61 training rows, and a quarter-end date is not a publication timestamp.

Reference and supervision. Original generated references are not independently established as fully grounded financial targets. All original DeepSeek train prompts imply an open think state without a supervised close marker. This retrospective retains that confound rather than calling parser normalization a target repair.

Audit boundary. Claude Opus5.5 High independently checked the matched 48 outputs, parser denominators, source bindings, all 1,230 Llama prepared targets and actual completed Llama training/selected adapter. Bounded followups corrected AI before/after descriptions while preserving raw outputs and human feedback. A subsequent independent review verified all 480 Llama outputs, reproduced the main results and reviewed the first three-model report, with no material defects. All four subsequent minor corrections and the final-test protocol were independently accepted. Claude subsequently accepted the actual 1,800-output run with 0 execution defects and identified two text defects plus a parser clarification. This revision applies D1-D3 using its verified facts; no repeat audit was requested. Corrected text was checked by Codex, not claimed as a new independent review. These reviews do not certify the complete assignment.

Reproducibility. Exact executed sources, source/model/tokenizer/adapter hashes, split and target receipts, environment and run configurations, selected adapters and raw outputs are retained. New course-named train_lora.py/train.sh/config.json/utils.py dispatchers describe future fresh reproductions; they were not the scripts used in the historical runs. Their default is a CPU plan, with no GPU execution or download.

Final test. All 1,800 outputs completed under the accepted frozen protocol after real human authorization. The original 20 previously inspected rows and the other 280 are reported separately; neither disclosure nor a future calendar split establishes a clean forecasting benchmark. Model/checkpoint/prompt/metric/serving choices were not selected from these test outputs.

Completion. This is a completed-experiments review artifact. Actual run evidence has passed independent review and its required wording corrections have been applied. The user accepted the review materials and authorized course submission. The submission README records the public Medium link and delivery status. The new matched-output AI judgments remain separate from actual human ratings.

Course source: frozen Fall-2026 Assignment1 FinGPT README, sections3-5. The dataset-generation step is optional; course raw scores require valid_count. Source URL: https://github.com/AI4Finance-Foundation/STAT-GR5398-Columbia-University/tree/master/Fall-2026/Assignment1/FinGPT